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The Carbon Dioxide Molecule: A New Derivation of the Potential, Spectroscopic, and Molecular Constants

ALAIN CHEDIN

Laboratoire de Météorologie Dynamique, Ecole Polytechnique, Route de Saclay, 91120 Palaiseau, France

AND

JEAN-LUC TEFFO

Laboratoire de Spectronomie Moléculaire, Université Pierre et Marie Curie, Tour 13,4 place Jussieu, 75230 Paris cedex 05, France

A new determination of the potential energy function of the carbon dioxide molecule from its vibrorotational spectrum is presented. Starting from the previous determination made by A. Chédin [*J. Mol. Spectrosc.* **76**, 430–491 (1979)] and from a significantly larger set of updated experimental data, the new potential is shown to provide a better agreement between theoretical and experimental eigenenergies. A similar improvement in the prediction of eigenstates is expected.

I. INTRODUCTION

In 1979, in a paper hereafter referred to as I (*1*), a first determination of the potential energy function of the carbon dioxide molecule from a very large set of vibrorotational data belonging to several isotopic species was presented. The theoretical support of this work was an extension of the algebraic approach previously developed for solving the eigenvalue problems (*2–5*), yielding the so-called "algebraized contact transformation," which is a particular case of the generalized Van Vleck transformation formalism (*6*). This study led to a significant improvement, up to the fourth order, of the potential previously derived only from vibrational energies, and provided calculated values of the spectroscopic and molecular constants for 10 isotopic species of CO_2.[1]

Since the completion of this work, numerous experimental results about the molecular constants of various isotopes of CO_2 were obtained from infrared and laser spectroscopy. Thus, the quantity and the accuracy of the present spectroscopic data led us to carry out a new numerical derivation of the potential. This is the purpose of this second paper. The new results, potential function and spectroscopic constants of four new isotopic species of CO_2 are reported here.

[1] Errata to paper I will be found in the Appendix.

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II. RESULTS

A. The Experimental Data Set

The data set in paper I gathered 214 band centers, 300 inertia constants, and 277 centrifugal distortion constants belonging to the first 10 isotopic species listed in Table I. This table shows what is now the experimental data distribution. We report in the last column the original references corresponding to the new constants introduced in the data set, (the source of the "old data" is indicated in the Appendix of paper I) together with a rough classification of the experimental technique involved. It can be seen that the general extension of the data set is about 20%. Of course, the modification of the set varies for each isotope. For $^{12}C^{16}O_2$, there are

TABLE I

Experimental Data Distribution

ISOTOPIC SPECIES	BAND CENTERS	INERTIA CONSTANTS	CENTRIFUGAL DIST.CONST.	NEW REFERENCES
$^{16}O^{12}C^{16}O$	96	124	140	a,b,b',c,d,e,f, g,h,i,s
$^{16}O^{13}C^{16}O$	49	64	73	a,b',c,d,e',j,k, l,l',m
$^{16}O^{12}C^{18}O$	36	46	48	a,c,d,l',n,n'
$^{16}O^{12}C^{17}O$	17	20	20	a,d,n,n'
$^{18}O^{12}C^{18}O$	22	33	35	c,l',o
$^{16}O^{13}C^{18}O$	10	10	9	l,m
$^{18}O^{13}C^{18}O$	9	13	13	c,e',q,r
$^{16}O^{13}C^{17}O$	5	7	5	q
$^{16}O^{14}C^{16}O$	3	4	4	c
$^{17}O^{12}C^{18}O$	3	3	3	p
$^{17}O^{12}C^{17}O$	3	2	2	p
$^{17}O^{13}C^{17}O$	1	3	0	q
$^{17}O^{13}C^{18}O$	1	3	0	q
$^{18}O^{14}C^{18}O$	0	3	3	c
TOTAL	255	335	355	

Laser spectroscopy : c(9), f(18), j(19), k(20), o(21)
F.T. spectroscopy (emission) : e(11), e'(12)
F.T. spectroscopy (absorption) : a(7), b(8), b'(28), d(10), g(13), h(14), i(15), l(16), l'(17), m(22), n(23), n'(24), p(25), q(26), r(27), s(29).

only six new levels, but many of the experimental constants were determined more accurately.

The last four isotopic varieties appear for the first time in the set. In addition, twice as many data are available for the $^{12}C^{18}O_2$ isotope, and we decided to add them to the four species (the first four in Table I) which participated in the previous fit of the potential.

B. The Potential Energy Function

The iterative process for the adjustment of the potential expanded with respect to the mass-independent quasinormal internal coordinates (the R's)[2] to the experimental data was described in paper I (Fig. 1). Starting from the solution found in paper I (Table II), a new series of iteration ended in the solution reported in the first column of Table II. The first column of Table III gives the coefficients of the expansion of the potential with respect to the dimensionless internal coordinates (the ξ's), while the expansions in terms of the R^*'s and ξ^*'s are, respectively, reported in the second columns of Tables II and III.

C. The Spectroscopic Constants

In Table IV we give the spectroscopic constants for the four new isotopic species included in the data set; $^{12}C^{17}O_2$, $^{13}C^{17}O_2$, $^{14}C^{18}O_2$, and $^{17}O^{13}C^{18}O$. Let us recall that these constants are the coefficients occurring in the expansions of the nonvanishing matrix elements of the transformed Hamiltonian (see Table I of paper I) in the quantum numbers v_1, v_2, l_2, v_3, and J. They are identified in the table by their dependences on the quantum numbers. For example, in the first column, v_1 means ω_1^0; v_2, ω_2^0; v_{1ll}, y_{1ll}; $v_3v_3v_3$, y_{333}, etc. The number following each constant is a code for the couplings involved (1 means "diagonal", i.e., no coupling). The second column gives the rotational diagonal spectroscopic constants: Here JJ should be read $[J(J+1) - l_2^2]$, so the first constant is B_0; v_1JJ, v_2JJ, and v_3JJ are the usual α^0's; $JJJJ$ is D_0; and the following constants are the γ's, the β's and H_0. The four remaining constants are denoted in paper I by a_{ll}, a_{1ll}, a_{3ll}, and a_{Jll}, and are, in fact, contributions to x_{ll}, y_{1ll}, y_{3ll}, and γ_{ll}. The third column gives the vibrational off-diagonal spectroscopic constants F_0, F_1, F_2, F_3, F_J, and F^{IV}. The numbers 2 and 11 refer to Fermi interactions of first and higher orders, respectively. The last column gives the rotational off-diagonal spectroscopic constants. The first five constants are L_0, L_1, L_2, L_3, and L_J, where L means l-type doubling constants (coupling number 5). The next six are the Coriolis interaction constants C_0, C_1, C_2, C_3, C_J, and C_l (coupling number 6), and the last is the coupling constant denoted in paper I by F^L, referring to Fermi + l-type interaction (coupling number 8).

[2] The various sets of coordinates used to expand the potential function are defined by Eqs. (4)–(7) of paper I.

TABLE II

Coefficients of the Expansion of the Potential Energy Function, W, with Respect to Quasinormal Internal Coordinates (in cm^{-1})

IDENT.[a]	W(R)	W(R*)
F 11	0.71007650D+03	0.71007727D+03
F 22	0.33547415D+03	0.33547392D+03
F 33	0.12315181D+04	0.12315194D+04
F 111	-0.49171774D+02	-0.14182262D+02
F 122	-0.29247846D+02	-0.29247842D+02
F 133	-0.26889675D+03	-0.76546356D+02
F 1111	0.20825738D+01	-0.25878799D+00
F 1122	0.13582064D+01	0.63780223D+00
F 1133	0.23106416D+02	-0.29209086D+01
F 2222	0.82616854D+00	0.82616740D+00
F 2233	-0.13179827D+01	-0.27798606D+01
F 3333	0.73924851D+01	-0.12405481D+01
F 11111	-0.21193008D-01	0.47436019D-01
F 11122	-0.40035943D-01	0.91365187D-02
F 11133	-0.66821333D+00	0.60846240D+00
F 12222	-0.11354627D+00	-0.11354617D+00
F 12233	-0.23216746D-01	-0.12538577D+00
F 13333	-0.11150774D+01	0.10111166D+00
F 111111	-0.18700266D-02	0.21151736D-02
F 111122	-0.83080756D-01	-0.84004111D-01
F 111133	-0.16800234D+00	-0.10287321D+00
F 112222	-0.81191552D-02	-0.89186900D-02
F 112233	-0.33872852D-01	-0.42680485D-01
F 113333	-0.15677234D+00	-0.14007476D+00
F 222222	0.74460718D-02	0.74480564D-02
F 222233	-0.53230633D-02	-0.10998372D-01
F 223333	0.19084759D-02	-0.90515525D-03
F 333333	-0.41278752D-02	-0.11543574D-01

[a] For example F_{12233} is the coefficient of $R_1R_2^2R_3^2$.

TABLE III

Coefficients of the Expansion of the Potential Energy Function, W, with Respect to Dimensionless Internal Coordinates (in cm^{-1})

IDENT.[a]	W(ξ)	W(ξ*)
f 11	0.54245356D+06	0.54245356D+06
f 13	0.84859079D+05	0.84859079D+05
f 22	0.19709107D+05	0.19709107D+05
f 111	-0.15188050D+07	-0.43389785D+06
f 113	-0.12510026D+06	-0.40241177D+05
f 122	-0.34871349D+05	-0.34871349D+05
f 1111	0.25898181D+07	-0.33923616D+06
f 1113	0.19427223D+06	0.28930797D+05
f 1122	0.17143568D+05	-0.17727781D+05
f 1133	0.83638623D+05	-0.81702811D+05
f 1223	0.97164983D+05	0.97164983D+05
f 2222	0.28515726D+04	0.28515726D+04
f 11111	-0.21359919D+07	0.12802648D+07
f 11113	0.13027503D+07	0.15951253D+07
f 11122	-0.25278366D+05	-0.25862579D+05
f 11133	-0.33396423D+06	-0.28285845D+06
f 11223	-0.53357170D+05	0.43807813D+05
f 12222	-0.79534511D+04	-0.79534511D+04
f 111111	-0.86111800D+07	-0.58687206D+07
f 111113	-0.88245834D+07	-0.86349065D+06
f 111122	-0.98986063D+06	-0.10489364D+07
f 111133	0.59684531D+07	0.66875013D+07
f 111223	-0.33300712D+07	-0.33396205D+07
f 111333	0.10574132D+08	0.95458322D+07
f 112222	-0.12428339D+05	-0.20381790D+05
f 112233	-0.46068853D+07	-0.48164347D+07
f 122223	-0.99370674D+04	-0.99370674D+04
f 222222	0.15099094D+04	0.15099094D+04

[a] For example f_{111223} is the coefficient of $(\xi_1^3\xi_3+\xi_1\xi_3^3)\xi_2^2$.

TABLE IV

Spectroscopic Constants (in cm^{-1}) for Four Isotopic Species of CO_2

(1)	VIBRATIONAL DIAGONAL (2)		(1)	ROTATIONAL DIAGONAL (2)		(1)	VIBRATIONAL COUPLINGS (2)		(1)	ROTATIONAL COUPLINGS (2)	
THE MOLECULE $^{17}O^{12}C^{17}O$ (CM-1)											
V_1	1	1.29599387D+03	JJ	1	3.67197789D-01	C	2	-2.53007524D+01	C	5	-1.36865901D-04
V_2	1	6.61873463D+02	V_1 JJ	1	1.10358160D-03	V_1	2	2.57311370D-01	V_1	5	5.23606276D-07
V_3	1	2.34293496D+03	V_2 JJ	1	-6.96147493D-04	V_2	2	3.19199132D-01	V_2	5	-3.96283787D-07
$V_1 V_1$	1	-2.78860489D+00	V_3 JJ	1	2.91584750D-03	V_3	2	1.09453941D-01	V_3	5	-3.27900259D-07
$V_1 V_2$	1	-5.09226025D+00	JJJJ	1	1.18023249D-07	JJ	2	1.17129855D-04	JJ	5	1.58183562D-10
$V_1 V_3$	1	-1.83902618D+01	$V_1 V_1$ JJ	1	3.77566324D-06	C	11	2.04369117D-02	C	6	-3.47689128D-02
L L	1	-1.00205212D+00	$V_1 V_2$ JJ	1	1.10228291D-05				V_1	6	3.00654446D-04
$V_2 V_2$	1	1.56210154D+00	$V_1 V_3$ JJ	1	1.03789785D-05				V_1	6	4.87881643D-04
$V_2 V_3$	1	-1.23925559D+01	L L JJ	1	5.54628971D-06				V_2	6	5.67861076D-05
$V_3 V_3$	1	-1.23459640D+01	$V_2 V_2$ JJ	1	-9.37095552D-06				V_3	6	-2.44189793D-09
$V_1 V_1 V_1$	1	1.58167705D-02	$V_2 V_3$ JJ	1	9.71010925D-06				L	6	5.05447435D-04
$V_1 V_1 V_2$	1	7.89038817D-03	$V_3 V_3$ JJ	1	5.66772206D-06				C	8	-8.62193029D-06
$V_1 V_1 V_3$	1	-6.49109402D-02	V_1 JJJJ	1	4.17204147D-11						
V_1 L L	1	6.39487519D-02	V_2 JJJJ	1	2.03383093D-09						
$V_1 V_2 V_2$	1	-5.38858809D-02	V_3 JJJJ	1	-2.57467503D-10						
$V_1 V_2 V_3$	1	8.89941037D-02	JJJJJJ	1	6.92507013D-15						
$V_1 V_3 V_3$	1	7.13676480D-02	L K	1	-1.90952936D-04						
V_2 L L	1	6.52376306D-03	V_1 L K	1	0.00000000D+00						
$V_2 V_2 V_2$	1	-3.88192561D-03	V_1 L K	1	-3.81905868D-04						
$V_2 V_2 V_3$	1	2.50612545D-02	V_3 L K JJ	1	9.00873043D-08						
$V_2 V_3 V_3$	1	-2.13387781D-02									
$V_3 V_3 V_3$	1	1.43113853D-02									
		5.19195012D-03									
THE MOLECULE $^{17}O^{13}C^{17}O$ (CM-1)											
V_1	1	1.29639423D+03	JJ	1	3.67215224D-01	C	2	-2.45806638D+01	C	5	-1.40940341D-04
V_2	1	6.42847857D+02	V_1 JJ	1	1.10414029D-03	V_1	2	2.44608725D-01	V_1	5	4.58694862D-07
V_3	1	2.27594762D+03	V_2 JJ	1	-6.60320469D-04	V_2	2	2.97714281D-01	V_2	5	-3.81923275D-07
$V_1 V_1$	1	-2.79638387D+00	V_3 JJ	1	2.80953594D-03	V_3	2	1.31346978D-01	V_3	5	-3.02172262D-07
$V_1 V_2$	1	-4.92862992D+00	JJJJ	1	1.17946409D-07	JJ	2	1.11801599D-04	JJ	5	1.63044312D-10
$V_1 V_3$	1	-1.79316258D+01	$V_1 V_1$ JJ	1	3.77566324D-06	C	11	1.83985359D-02	C	6	-3.36327032D-02
L L	1	-9.43515081D-01	$V_1 V_2$ JJ	1	1.05867235D-05				V_1	6	2.54930218D-04
$V_2 V_2$	1	1.47174142D+00	$V_1 V_3$ JJ	1	1.01338160D-05				V_1	6	4.64836952D-04
$V_2 V_3$	1	-1.16074147D+01	L L JJ	1	5.12596801D-06				V_2	6	5.88251193D-05
$V_3 V_3$	1	-1.15797245D+01	$V_2 V_2$ JJ	1	-8.81669907D-06				V_3	6	-3.22961796D-09
$V_1 V_1 V_1$	1	1.58167705D-02	$V_2 V_3$ JJ	1	9.12739788D-06				L	6	4.79002482D-04
$V_1 V_1 V_2$	1	2.13649846D-04	$V_3 V_3$ JJ	1	5.29225716D-06				C	8	-8.81911541D-06
$V_1 V_1 V_3$	1	-6.51154284D-02	V_1 JJJJ	1	4.17204147D-11						
V_1 L L	1	5.61909083D-02	V_2 JJJJ	1	1.95610835D-09						
$V_1 V_2 V_2$	1	-4.66882990D-02	V_3 JJJJ	1	-2.55702925D-10						
$V_1 V_2 V_3$	1	8.24785098D-02	JJJJJJ	1	6.92507013D-15						
$V_1 V_3 V_3$	1	7.09835148D-02	L K	1	-1.96642631D-04						
V_2 L L	1	5.88088122D-03	V_1 L K	1	0.00000000D+00						
$V_2 V_2 V_2$	1	-3.46179091D-03	V_1 L K	1	-3.93285258D-04						
$V_2 V_2 V_3$	1	2.39390341D-02	V_3 L K JJ	1	9.55358264D-08						
$V_2 V_3 V_3$	1	-2.08588949D-02									
$V_3 V_3 V_3$	1	1.33134819D-02									
		4.70507533D-03									

(1) IDENTIFICATION OF THE CONSTANT (2) COUPLING NUMBER

TABLE IV—*Continued.*

VIBRATIONAL DIAGONAL (1)	(2)		ROTATIONAL DIAGONAL (1)	(2)		VIBRATIONAL COUPLINGS (1)	(2)		ROTATIONAL COUPLINGS (1)	(2)	
THE MOLECULE $^{18}O^{14}C^{18}O$ (CM-1)											
V_1	1	1.26040362D+03	JJ	1	3.468509140-01	C	2	-2.34257448D+01	C	5	-1.30142900D-04
V_2	1	6.21100242D+02	V_1 JJ	1	1.01401781D-03	V_1	2	2.25381131D-01	V_1	5	3.96966154D-07
V_3	1	2.19939145D+03	V_2 JJ	1	-5.99483123D-04	V_2	2	2.73199295D-01	V_2	5	-3.37886409D-07
$V_1 V_1$	1	-2.64214827D+00	V_3 JJ	1	2.55915733D-03	V_3	2	1.26227126D-01	V_3	5	-2.64724027D-07
$V_1 V_2$	1	-4.62197417D+00	JJJJ	1	1.05144343D-07	JJ	2	1.00177229D-04	JJ	5	1.42226023D-10
$V_1 V_3$	1	-1.68494460D+01	$V_1 V_1$ JJ	1	3.36777201D-06	C	11	1.65175083D-02	C	6	-3.11103336D-02
L L	1	-8.81178439D-01	$V_1 V_2$ JJ	1	9.36250963D-06				V_1	6	2.22510710D-04
$V_2 V_2$	1	1.37397412D+00	$V_1 V_3$ JJ	1	8.99472612D-06				V_2	6	4.16339280D-04
$V_2 V_3$	1	-1.08139955D+01	L L JJ	1	4.49704570D-06				V_3	6	5.36970353D-05
$V_3 V_3$	1	-1.07925051D+01	$V_2 V_2$ JJ	1	-7.76483320D-06				JJ	6	-2.98392686D-09
$V_1 V_1 V_1$	1	1.45171006D-02	$V_2 V_3$ JJ	1	8.03693171D-06				L	6	4.28592615D-04
$V_1 V_1 V_2$	1	-1.13221156D-03	$V_3 V_3$ JJ	1	4.65307084D-06				C	8	-8.01757674D-06
$V_1 V_1 V_3$	1	-5.98257598D-02	V_1 JJJJ	1	3.61647449D-11						
$V_1 L L$	1	5.02180638D-02	V_2 JJJJ	1	1.68179424D-09						
$V_1 V_2 V_2$	1	-4.16004431D-02	V_3 JJJJ	1	-2.21281324D-10						
$V_1 V_2 V_3$	1	7.45249530D-02	JJJJJJ	1	5.83375686D-15						
$V_1 V_3 V_3$	1	6.51096525D-02	L K	1	-1.81589926D-04						
$V_2 L L$	1	5.28233116D-03	V_1 L K	1	0.00000000D+00						
$V_2 V_2 V_2$	1	-3.10232282D-03	V_3 L K	1	-3.63179848D-04						
$V_3 L L$	1	2.17659628D-02	L K JJ	1	8.62620347D-08						
$V_2 V_3 V_3$	1	-1.90542590D-02									
$V_2 V_3 V_3$	1	1.20375231D-02									
$V_3 V_3 V_3$	1	4.22939274D-03									
THE MOLECULE $^{17}O^{13}C^{18}O$ (CM-1)											
V_1	1	1.27815466D+03	JJ	1	3.56947184D-01	C	2	-2.43116854D+01	C	5	-1.33672403D-04
V_2	1	6.40404365D+02	V_1 JJ	1	1.05842486D-03	V_1	2	2.40241924D-01	V_1	5	4.55664392D-07
V_3	1	2.26748192D+03	V_2 JJ	1	-6.44850764D-04	V_2	2	2.94417516D-01	V_1	5	-3.65736433D-07
$V_1 V_1$	1	-2.71517310D+00	V_3 JJ	1	2.72778081D-03	V_3	2	1.19259301D-01	V_2	5	-2.94769413D-07
$V_1 V_2$	1	-4.84409334D+00	JJJJ	1	1.11442866D-07	JJ	2	1.08141023D-04	JJ	5	1.50260384D-10
$V_1 V_3$	1	-1.75817119D+01	$V_1 V_1$ JJ	1	3.56278133D-06	C	11	1.82918522D-02	C	6	-3.28505202D-02
L L	1	-9.36681276D-01	$V_1 V_2$ JJ	1	1.01364683D-05				V_1	6	2.57746178D-04
$V_2 V_2$	1	1.46085720D+00	$V_1 V_3$ JJ	1	9.68172671D-06				V_2	6	4.49935701D-04
$V_2 V_3$	1	-1.15446153D+01	L L JJ	1	4.97858218D-06				V_3	6	5.52234168D-05
$V_3 V_3$	1	-1.15190170D+01	$V_2 V_2$ JJ	1	-8.50979713D-06				JJ	6	-2.75093706D-09
$V_1 V_1 V_1$	1	1.51305343D-02	$V_2 V_3$ JJ	1	8.82120023D-06				L	6	4.64371754D-04
$V_1 V_1 V_2$	1	2.55408495D-03	$V_3 V_3$ JJ	1	5.11004939D-06				C	8	-8.32310567D-06
$V_1 V_1 V_3$	1	-6.21321539D-02	V_1 JJJJ	1	3.95110865D-11						
$V_1 L L$	1	5.63221163D-02	V_2 JJJJ	1	1.84727138D-09						
$V_1 V_2 V_2$	1	-4.70236900D-02	V_3 JJJJ	1	-2.39798101D-10						
$V_1 V_2 V_3$	1	8.13763416D-02	JJJJJJ	1	6.34054914D-15						
$V_1 V_3 V_3$	1	6.79645899D-02	L K	1	-1.86504705D-04						
$V_2 L L$	1	5.84890048D-03	V_1 L K	1	-4.08566394D-08						
$V_2 V_2 V_2$	1	-3.45747265D-03	V_3 L K	1	-3.72968549D-04						
$V_3 L L$	1	2.32282483D-02	L K JJ	1	8.84219602D-08						
$V_2 V_2 V_3$	1	-2.00609488D-02									
$V_2 V_3 V_3$	1	1.29660295D-02									
$V_3 V_3 V_3$	1	4.72129699D-03									

D. The Molecular Constants

From the updated solution obtained for the potential we have computed the molecular constants ν_0, B, and D of levels belonging to 14 isotopic species of carbon dioxide. The results are summarized in Table V, which provides a statistical comparison between theory and experiment. It can be seen that the agreement obtained here represents a significant improvement as compared to paper I, except for the centrifugal distortion constants. This may be explained by the fact that the extension of the experimental data set is much more important for the D's (almost 100 items), while the accuracy of the data is obviously less than it is for the ν_0's and the B's.

This improvement may also be illustrated by an example. In paper I a slight discrepancy was observed between calculated and experimental values of B associated with the $(100,02^00)_{I,II}$ and 001 states of several symmetric species of CO_2. Here, this discrepancy is for the most part resolved, since the B values associated with the 001 states are in agreement with the laser data of Ref. (*9*) within 2×10^{-6} cm^{-1} for six isotopic species. The agreement for the ΔB values of the laser transition is about 5×10^{-6} cm^{-1}.

III. CONCLUSION

The present work provides an updating of the results previously given in paper I, since we obtained a better agreement between theory and experiment with a significantly larger experimental data set, including data for four new isotopic species.

TABLE V

Mean Deviations from Theory to Experiment

1979	BAND CENTERS	INERTIA CONSTANTS	CENT. DIST. CONSTANTS
NUMBER OF DATA	214	300	277
MEAN DEVIATION	0.035	2.2×10^{-5}	3.0×10^{-9}
1983			
NUMBER OF DATA	255	335	355
MEAN DEVIATION	0.033	2.0×10^{-5}	3.4×10^{-9}

Since paper I was published, Lacy (*30*) determined the sextic force field of CO_2 by direct diagonalization of the rotation–vibration Hamiltonian matrix. The potential was calculated using vibrational frequencies and B values pertaining to the main isotopic species (information provided by the ground state D constant was also included during the fit). From the comparison of these results with those of Table I (Table II of Ref. *30*)), Lacy concluded that the contact transformation formalism provides a more accurate prediction of molecular constants.

Furthermore, we believe that the present potential allows not only a good determination of the eigenenergies but also a convenient prediction of the eigenstates. This was verified with the solution reported in paper I, as well in infrared (*31*) as in Raman (*32*) intensity calculations.

APPENDIX: ERRATA TO PAPER I

A. Chédin, J. Mol. Spectrosc. ***76, 430–491 (1979)***

1. On page 436, Table I, last column, The matrix element number 11 participates in the computations and "NO" should be changed to "YES." On the contrary the matrix element number 12 does not participate in the computations: "YES" should be changed to "NO." As a consequence, the identification number "12" should be changed to "11" on pages 436 (line 8 below Table I and in Eq. (18)) and 444 (line 15 below Table VIII).

2. On pages 447–456, Table IX, second column, the signs for $v_2JJ(\alpha_2^0)$ and $v_3JJJJ(\beta_3)$ should be changed from positive to negative.

3. On pages 447–456, Table IX, third column, last line, "12" should be changed to "11." The values of the corresponding constants (identified by "C" in the table and by "F^{IV}" in the text) should be replaced by the following values:

$^{16}O^{12}C^{16}O$	$2.04440887 \times 10^{-2}$
$^{16}O^{13}C^{16}O$	$1.84049806 \times 10^{-2}$
$^{18}O^{12}C^{18}O$	$1.99961221 \times 10^{-2}$
$^{18}O^{13}C^{18}O$	$1.79955101 \times 10^{-2}$
$^{16}C^{14}C^{16}O$	$1.67112722 \times 10^{-2}$
$^{16}O^{12}C^{18}O$	$2.01994125 \times 10^{-2}$
$^{16}O^{12}C^{17}O$	$2.03208232 \times 10^{-2}$
$^{16}O^{13}C^{18}O$	$1.81699415 \times 10^{-2}$
$^{16}O^{13}C^{17}O$	$1.82887484 \times 10^{-2}$
$^{17}O^{12}C^{18}O$	$2.00999759 \times 10^{-2}$

4. On page 442, Table VI, the coefficient of $W(\xi^*)$, f_{111}, should read "$-0.4356592E$ + 06" instead of "$-9.4356592E$ + 06." The coefficients of $W(\xi)$, f_{1111} and f_{2222}, should read, "$0.2591949E$ + 07" instead of "$0.2491949E$ + 07," and "$0.2842204E$ + 04" instead of "$0.2844204E$ + 04."

Fortunately, these errors are only *editing errors*. All the results presented in this paper (potential, spectroscopic, and molecular constants, vibrational or transition energies, etc.) are correct.

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